3-Finding Floor Value

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer:(penalty regime: 0 %)

```
#include <stdio.h>

int findFloor(int arr[], int low, int high, int x) {
  if (low > high)
    return -1;

  int mid = low + (high - low) / 2;

  if (arr[mid] == x)
    return arr[mid];

  if (arr[mid] > x)
    return findFloor(arr, low, mid - 1, x);

  int result = findFloor(arr, mid + 1, high, x);

  if (result != -1)
    return result;

  return arr[mid];
}

int main() {
```

```
    int n;
    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    int x;
    scanf("%d", &x);

    printf("%d\n", findFloor(arr, 0, n - 1, x));

    return 0;
}
```

Feedback

	Input	Expected	Got	
	6 1 2 8 10 12 19 5	2	2	
	5	85	85	

	Input	Expected	Got	
	10 22 85 108 129 100			
	7 3 5 7 9 11 13 15 10	9	9	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.